Python data Structures:

→ list
→ tuples
→ dictionaries
→ Sets

everything in python is treated as an object.

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list → changeable, zero-based indexing,

Nested list

```
[
 ['J', 'Q', 'K'],
 ['2', '2', '2'],
  ['6', 'A', 'K'],
    ]
```

list = ['AI', 23.07, "Seventy"]

Planets[-1] → gives last item

Planets[-2] → gives Second last item

Slicing → [start:stop]

[0:3] → Start from 0 and go till 3. 0, 1, 2.

[:3] → give me everything from start till item 3. (0, 1, 2)

[3:] → Start at item3 and give me everything else

Planets[1:-1] → Start from Second position and Stop right before last one

marks = [3, 5, 6, "Harry", True]

marks[-3] → marks[len(marks) − 3]

= 5-3 → [2]

marks[2] → 6

0 1 2 3 4

[Mercury, Venus, Earth, Mars, Neptune]

[1:-1] → [1:4] → 1, 2, 3

(list functions)

l.append → adds an element at end.

l.sort → Ascending order

Python Tuples:. → t = (1, 2, 3)
→ they cannot be changed | Immutable

t [0] = 100 (x) error.

Tuple functions:.

→ count ()
↓
counts the no of instances

→ index()
↓
tells the first position (index) of where value is found

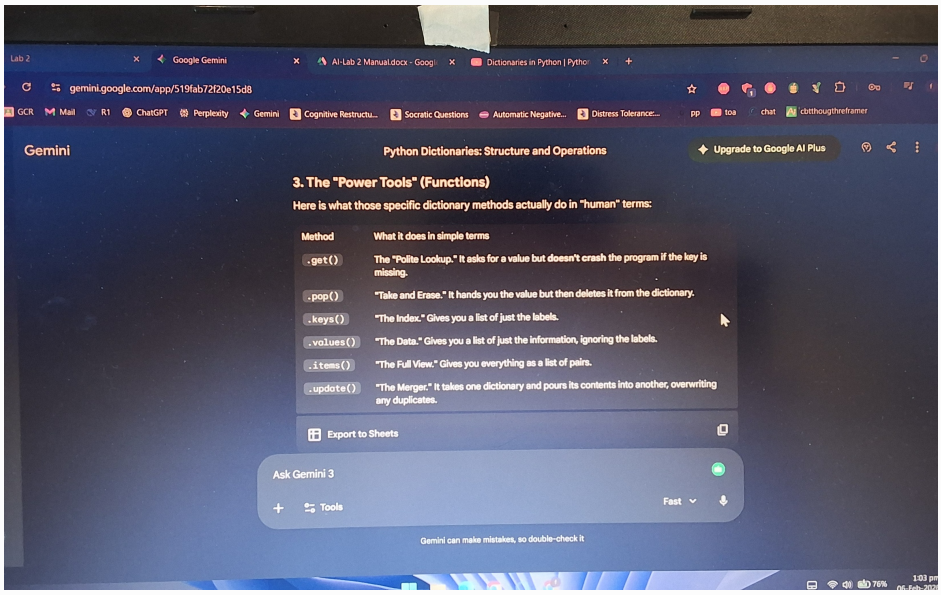
Dictionaries :.

dic = {
"Harry" : "Human being",
"Spoon" : "Object"
}

Print (dic ["Harry"])
↑ key

→ Humanbeing
↓
value printed

Duplicate keys are not allowed !! and its key is immutable



Accessing multiple all keys:

print(info.keys()) print(info.values())

↓

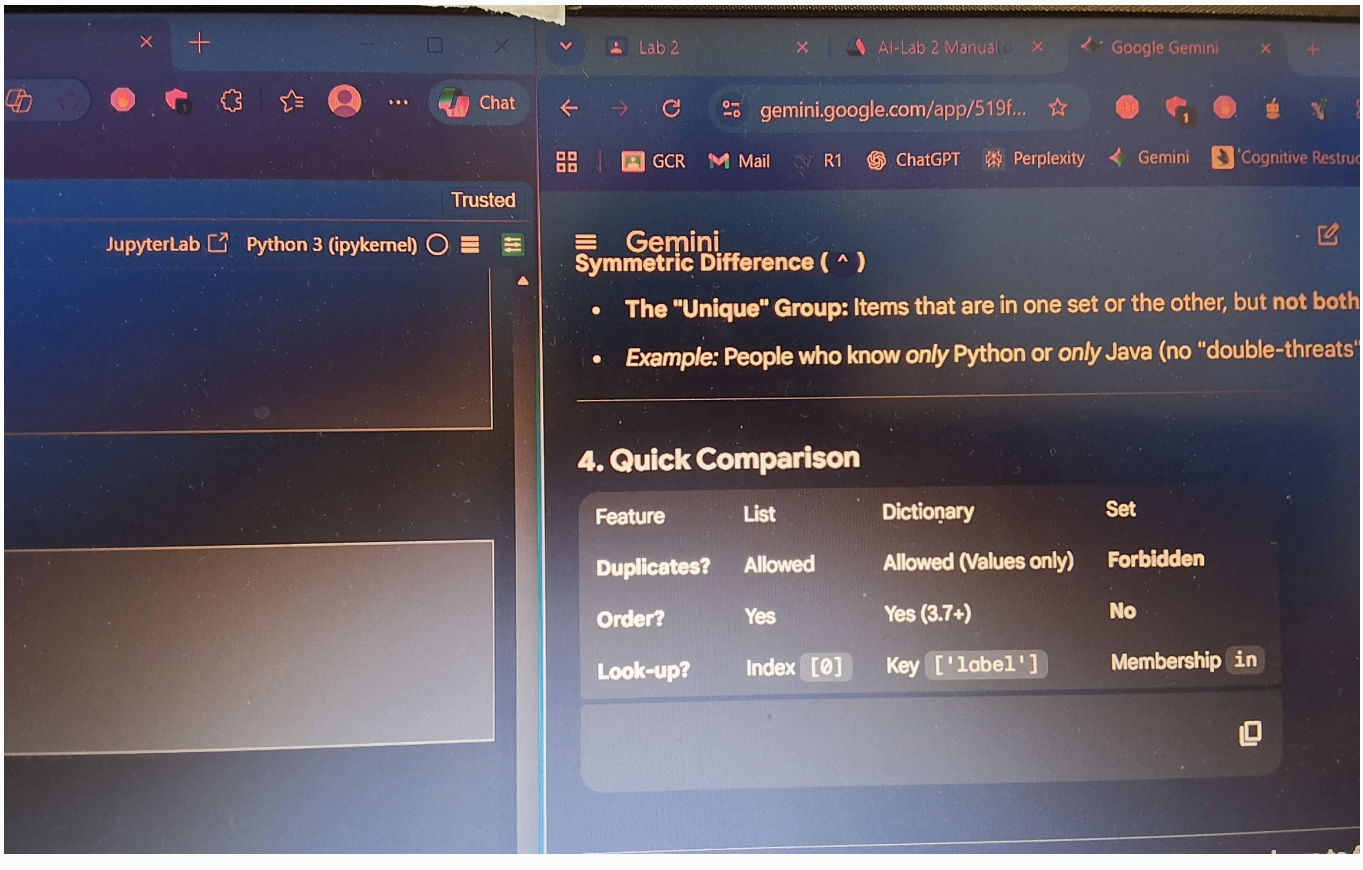
name of dictionaries

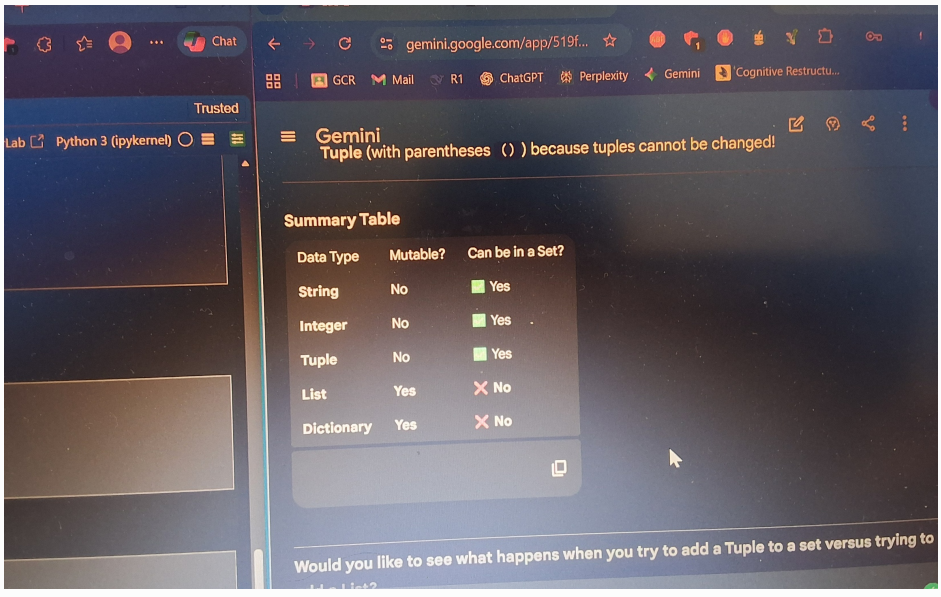
Set → only thing that matters is that is this in the Bag or not.

→ no duplicates

→ unordered

→ we can place only hard things in the bag (like Strings, numbers, tuples etc) but not mutable things like list.




Chat
gemini.google.com/app/519f...
GCR
Mail
R1
ChatGPT
Perplexity
Gemini
'Cognitive Restructu...
Trusted
Lab
Python 3 (ipykernel)
Gemini
Tuple (with parentheses ( )) because tuples cannot be changed!
Summary Table
Data Type
Mutable?
Can be in a Set?
String
No
Yes
Integer
No
Yes
Tuple
No
Yes
List
Yes
No
Dictionary
Yes
No
Would you like to see what happens when you try to add a Tuple to a set versus trying to